Memory Footprints from Resident Memory and Allocator Statistics: A Footprint Model at the Cost of a Native Run

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Abstract

A memory-aware scheduler needs to know how much memory a job uses over its run. MemPrint, a Pin-based footprint profiler, measures the pages a program touches, but its windowed mode costs 12.0× a native run at the median of 22 programs, 18.7× to 93.0× on programs that allocate and free memory often, and up to 67.4× on a language model engine. The resident set size (RSS) of a native run costs nothing extra, but it overestimates allocator-heavy programs by a median of 16% over the whole run, and miniVite by 102%, because the allocator keeps freed memory resident and RSS counts shared libraries. We propose a footprint model that samples, from one thread inside the process every 5 ms, the anonymous RSS, the proportional set size, the RSS of mapped data files, and the free bytes the allocator reports (glibc, jemalloc or TCMalloc). Its data estimate, anonymous RSS minus free bytes plus mapped data files and the runtime's pages, has a median whole-run error of −4% on the 7 allocator-heavy programs of a 23-program corpus (−7% on miniVite), against −5% for MemPrint, at a median of 0.99× the run time of a plain RSS run. Transferred from smaller inputs to a held-out larger one, it has a median error of −4%, against +20% for RSS and −6% for MemPrint. In a cluster scheduling simulator that reserves memory by each job's estimated curve, it gives the lowest mean response time of all estimates that profile the job's own input (433 minutes at a load of 0.8, against 515 for the RSS curve and 519 for MemPrint). Under TCMalloc its median error is −4%. Under jemalloc it is −1% with transparent huge pages disabled and +35% with them enabled, because the kernel backs memory the program touches in part with whole 2 MiB pages. On llama2.c and llama.cpp, the data-file term brings the model from between −39% and −92% to between −28% and +15% of the truth.

1 Introduction

A scheduler that packs jobs onto nodes by their memory use needs an estimate of each job's footprint over time, because a job's peak lasts only part of its run [2, 20, 11]. A reservation that follows the footprint over time lets a node hold more jobs than a reservation of the peak, but the estimate must not fall below what the job uses, or the job runs out of memory.

MemPrint measures a program's footprint with Pin [13]. It records the bytes and pages each program touches and the time at which each block is freed. Its windowed mode watches 5% of the run and costs 12.0× a native run at the median on 22 programs of our corpus. However, on the programs where a footprint estimate helps the most, those that allocate and free memory throughout their run, the cost rises to 18.7× to 93.0×, because Pin instruments every memory access and every allocator call.

The resident set size of a native run is the obvious cheap alternative. On array programs, RSS is the footprint, because these programs allocate their arrays once and touch them. On allocator-

heavy programs, RSS overestimates the footprint over the run by a median of 16%, and by 102% on miniVite [7], because the kernel counts memory that the program no longer uses.

In this work, we make the following contributions:

- We identify the parts of RSS that are not part of the footprint, freed memory that the allocator keeps resident and the pages of shared libraries, and we measure their size per program (Section 5.1).
- We propose a footprint model that combines RSS, PSS, the RSS of mapped data files and the allocator's free bytes, sampled every 5 ms from a thread inside the process through an `LD_PRELOAD` library (Section 3).
- We show on 23 programs that the model matches MemPrint's accuracy on the job's own input and on larger inputs predicted from smaller ones, at a median of 0.99× the cost of a plain RSS run (Section 5 and Section 6).
- We show in a cluster scheduling simulator that the model's curve gives the lowest response time of all estimates of the job's own input (Section 7).
- We evaluate the model under jemalloc and TCMalloc and on two language model engines, and we show that transparent huge pages are the remaining source of disagreement with a 4 KiB pages-touched truth (Section 8 and Section 9).
- We test four other ways to correct RSS, each alone, and show why each is less accurate (Section 10).

2 Background: what RSS counts

The RSS of a process is the number of its pages in memory. An anonymous page enters RSS when the program first touches it and leaves when the program or its allocator unmaps it or releases it with `madvise`. A file-backed page, such as a page of a shared library's code or of a mapped data file, enters RSS when any process reads it and stays as long as it is mapped and the kernel does not reclaim it. The *footprint* we estimate is the set of pages the program has touched and not yet freed, plus the pages the runtime needs before the program starts. RSS differs from the footprint in five ways.

Allocator retention. The C library does not return every freed block to the kernel. glibc's `malloc` serves blocks below a threshold (128 KiB, raised dynamically up to 32 MiB) from its arenas, keeps freed blocks in free lists for reuse, and releases memory only from the top of the heap when the free space there exceeds the trim threshold [6]. jemalloc and TCMalloc also release freed memory late or in batches [5, 8]. A program that frees and allocates memory throughout its run therefore holds freed pages in RSS. Each of these allocators reports the bytes it holds free through its own interface.

Shared libraries. RSS counts every resident page of every mapped file, including the code of the C library, the C++ runtime and, for MPI programs, the MPI library. These pages do not depend on the input, and other processes on the node map the same pages. The PSS of a process divides each resident page by the number of processes that map it.

Mapped data files. A program that maps its input, such as a language model engine that maps its weights, holds that input in file-backed pages. These pages are part of the footprint, unlike the pages of libraries.

Language runtimes with their own allocators. CPython's object allocator takes 256 KiB arenas from the system and returns an arena only when all its objects are freed. GCC's garbage-collected heap also maps its own pages. The C library's statistics do not cover these allocators.

Page size. RSS counts whole pages. On our corpus, the 4 KiB pages a program touches exceed the bytes it touches by 1% at the median, so we use the pages as the truth. With transparent huge pages [12], the kernel can back a 2 MiB aligned range of anonymous memory with one huge page at the first touch, and RSS then counts 2 MiB even if the program touches only part of the range [10, 17]. Our machine enables transparent huge pages for all anonymous memory (the `always` setting).

3 The footprint model

The model estimates a footprint curve from five quantities sampled at the same time inside the process. Let $A(t)$ be the anonymous RSS, $P(t)$ the PSS, $M(t)$ the RSS of mapped data files, $F(t)$ the bytes the allocator holds free, and B the runtime's own pages, the smallest RSS of an earlier run of the program, which is measured once per program. The model gives two estimates.

Estimate 1, the data footprint, is

$$\hat{f}_1(t) = \max\big(A(t) - F(t),\, 0\big) + M(t) + B. \tag{1}$$

It counts the anonymous pages the allocator has not freed and the pages of mapped data files, and it leaves out the pages of libraries. It estimates the memory that grows with the input, which is what a scheduler predicts and what our truth measures.

Estimate 2, the charged footprint, is

$$\hat{f}_2(t) = \max\big(P(t) - F(t),\, 0\big). \tag{2}$$

PSS is the anonymous RSS plus each file-backed page divided by the number of processes that map it, so the estimate charges the job for its share of shared libraries and data files and for all of a library that only it maps. It estimates the memory the job adds to a node.

Both estimates subtract free bytes from resident memory rather than count live allocations. A counter of live `malloc` blocks misses every allocation that does not go through `malloc`, so an estimate built from it falls below the truth whenever a program uses its own allocator. Subtracting free bytes errs the other way. Memory that the allocator does not report stays in RSS, so a program whose allocator the model cannot see gets the RSS estimate, which is too high but safe.

Free bytes. The library looks up the allocator's interface when it starts. With jemalloc (`mallctl`), the free bytes are `stats.resident` minus `stats.allocated` minus `stats.metadata`, after a refresh of the statistics through `epoch`. With TCMalloc (`MallocExtension_GetNumericProperty`), they are `generic.heap_size` minus `tcmalloc.pageheap_unmapped_bytes` minus `generic.current_allocated_bytes`. Otherwise they are glibc's `mallinfo` field `fordblks`, the free bytes of all its arenas. The machine runs glibc 2.28, which has no `mallinfo2`, so glibc's free bytes are a 32-bit field that wraps above 4 GiB. A newer glibc reports them in 64 bits.

Mapped data files. A data file is a file mapping that is not the executable, a shared object (a path that contains `.so`), or the C library's locale and character set data. $M(t)$ is the sum of the RSS of these mappings.

Implementation. A shared library of 162 lines of C, loaded with `LD_PRELOAD`, starts one thread in its constructor. Every 5 ms the thread reads the process's memory counters, queries the allocator,

and writes one line to a buffered file. It has two modes. The first reads `/proc/self/smaps_rollup`, which gives A, P and RSS but not M. The second reads `/proc/self/smaps`, one record per mapping, which also gives M. On the 23 corpus programs, M is at most 0.07 MiB, so we evaluate the corpus with the first mode and the language model engines with the second. The thread does not stop the program's threads, and it takes the allocator's locks only for the duration of its query. The library removes itself from `LD_PRELOAD` so that child processes are not sampled, and a program made of several processes would need one output file per process.

4 Methodology

Machine. All runs use a two-socket Intel Xeon E5-2620 v4 (16 cores, 32 threads) with 62 GiB of memory, Linux 4.18 with transparent huge pages enabled, and glibc 2.28. Runs whose time we report execute one at a time on an otherwise idle machine.

Corpus. We use the 23 programs of the corpus for which a pages-touched truth exists, each at its largest input: PolyBench (2mm, atax, gemm, jacobi-2d), the NAS parallel benchmarks (cg, is, mg), the GAP benchmark suite (bfs, cc, pr, sssp) [3], the proxy applications HPCCG, Kripke, LULESH, miniFE, miniVite and Quicksilver, darknet AlexNet inference, GCC's `cc1` compiling SQLite, and Lua [9], CPython, Perl and SQLite running table, JSON, hash and SQL workloads. We call 7 programs *allocator-heavy*, because they allocate and free memory throughout their run: miniVite, CPython, Lua, SQLite, Perl, Quicksilver and `cc1`. We chose the set before these experiments, from the earlier observation that RSS and MemPrint disagree on these programs and agree on the others. The runs last 0.1 to 12 s and peak at 27 to 273 MiB of RSS. Every program except darknet also has one or two smaller inputs. Separately, we use llama2.c with three model sizes (15M, 42M and 110M parameters) and llama.cpp with a 135M-parameter model generating 32 and 128 tokens.

Truth. The truth is the number of 4 KiB pages the program touches and has not freed, which Pin measures in MemPrint's splitter mode with page counting and free tracking, plus the runtime's pages B. Pin's time axis is its count of memory references. We place the truth on the wall clock of a native run through hardware counts of the loads and stores that the native run executes in every 5 ms interval. The truth at a share f of Pin's references is placed at the time at which the native run had executed a share f of its references. We divide each run into 100 equal steps of progress and take the largest value of each curve in each step. The estimates are placed on the same 100 steps by the wall clock of their own run. The truths under jemalloc and TCMalloc come from Pin runs with the same allocator, and they are placed on the wall clock by the counts of the native run with glibc.

Metrics. We report three errors of an estimate e against the truth t over the 100 steps. The signed whole-run error is $(\sum e - \sum t)/\sum t$, which is positive when the estimate reserves more memory over the run than the program uses. The absolute whole-run error is $\sum |e - t|/\sum t$. The peak error is $\max e/\max t - 1$. Each program runs 3 times, and we report the median of each error over the runs. The overhead of an estimate is the median wall time of its run over the median wall time of a run under the plain RSS sampler of the corpus (`rss_sampler.py`, an external process that reads `smaps_rollup` every 5 ms and takes 40 ms to start).

Baselines. *RSS* is the RSS curve of the plain sampler's run. *MemPrint* is the windowed mode that watches 5% of the run (W5), placed on the native wall clock by the same reference counts as the truth, plus B. Because the truth and MemPrint pass through the same clock mapping, MemPrint's timing error against the truth is smaller than its timing error against the real run. Its cost is the wall time of the W5 run and of the counter run over the wall time of an earlier plain

Table 1: Medians over programs, each the median of 3 runs. Whole-run errors and peak errors are percentages of the truth (pages touched plus the runtime's pages, on the native clock). The 90th percentile of the absolute error and the lowest peak error show the worst cases. Overhead is the run time over a plain RSS run (×).

			Whole-run error (%)			Peak error (%)		Overhead
Programs	Estimate	n	signed	absolute	q90 abs.	median	lowest	(×)
Allocator-heavy (7)	RSS	7	+16	17	55	+10	0	1.00
	Data (1)	7	−4	8	10	−2	−4	0.99
	Shared (2)	7	0	9	23	−1	−3	0.99
	MemPrint	7	−5	5	17	−5	−16	34.4
Others (16)	RSS	16	+1	2	8	0	−4	1.00
	Data (1)	16	−1	3	8	−1	−5	0.98
	Shared (2)	16	−5	7	10	−4	−9	0.98
	MemPrint	15	−1	1	3	−1	−2	8.9
All (23)	RSS	23	+4	7	20	+1	−4	1.00
	Data (1)	23	−1	5	9	−1	−5	0.99
	Shared (2)	23	−4	8	12	−3	−9	0.99
	MemPrint	22	−1	2	12	−1	−16	12.0

RSS run. MemPrint has no W5 curve for darknet, so its rows cover 22 programs.

5 Accuracy on the job's own input

Allocator-heavy programs. The data estimate reduces the median whole-run error of RSS from +16% to −4% and the median absolute error from 17% to 8% (Table 1). MemPrint has a median signed error of −5% and a median absolute error of 5%. Per program (Table 2), the data estimate reduces miniVite's error from +102% to −7%, CPython's from +25% to −3%, and Lua's from +20% to −1%, and its absolute error is at most 11% on every allocator-heavy program. Its peak is within 4% of the truth's on all 7 programs, with a median of −2%, while the RSS peak is 10% too high at the median. MemPrint's lowest peak error is −16%, on Lua.

The charged estimate. The charged estimate has a median signed error of 0% on the allocator-heavy programs, and it lies above the truth on miniVite (+37%), `cc1` (+12%) and Quicksilver (+8%), because it keeps the pages of libraries that only this job maps. Unlike the data estimate, the charged estimate counts memory that the truth does not count, so its error against the truth does not measure its accuracy as an estimate of a node's memory use.

Other programs. On the 16 other programs, RSS is already accurate, with a median signed error of +1% and a median absolute error of 2%. The data estimate has −1% and 3%, and MemPrint has −1% and 1%.

The truth's clock. An earlier version of this comparison kept the truth on Pin's count of memory references. Every estimate then had whole-run errors of 10% or more in the same direction on 7 of the 16 other programs, from +10% to +35% on the GAP programs and from −10% to −25% on atax, NPB-IS and NPB-MG, with peak errors within 6%. A phase that executes few memory references per second, such as GAP's graph construction, takes a smaller share of the run on Pin's clock than on the wall clock. Placing the truth on the native wall clock removes these errors, and the wall clock is the axis on which a scheduler observes a job.

Overhead. The model's run takes 0.99× the time of a plain RSS run at the median, and 0.96×

Table 2: Whole-run error per program (% of the truth), signed and absolute, median of 3 runs. Overhead is the run time over a plain RSS run (×). The model's two estimates come from the same run. MemPrint has no W5 run of darknet.

	RSS		Data (1)		Shared (2)		MemPrint		Overhead (×)	
Program	sign.	abs.	sign.	abs.	sign.	abs.	sign.	abs.	model	MemPrint
Allocator-heavy										
miniVite	+102	102	−7	9	+37	38	−19	19	0.99	40.2
python	+25	25	−3	8	0	5	−3	3	1.00	27.3
lua	+20	21	−1	5	−4	7	−16	16	0.98	23.6
sqlite	+9	13	+1	11	0	11	+1	4	0.96	50.3
perl	+11	11	−9	9	−8	9	−13	13	0.99	34.4
quicksilver	+16	17	−4	4	+8	9	−5	5	0.99	18.7
cc1	+15	15	−5	5	+12	12	−4	4	1.00	93.0
Others										
2mm	−1	2	0	1	−4	4	0	0	1.01	5.9
atax	−6	7	−6	7	−12	12	−1	1	0.60	23.1
darknet	+1	1	−8	9	−9	9	–	–	1.21	–
gap-bfs	+8	10	+3	7	−1	9	−3	3	0.98	8.9
gap-cc	+5	7	−1	5	−5	8	−3	3	0.96	8.5
gap-pr	+5	7	−1	6	−5	9	−3	3	0.97	7.4
gap-sssp	+4	7	+5	9	+3	10	−3	3	1.04	8.3
gemm	−1	1	−1	1	−5	5	−1	1	1.02	5.6
hpccg	0	1	−1	1	−3	3	−1	1	0.98	12.2
jacobi-2d	−2	2	−1	1	−6	6	−1	1	1.00	4.4
kripke	+1	2	0	1	0	1	0	0	1.01	11.5
lulesh	+6	9	+4	7	+2	8	−1	1	1.00	11.3
minife	+3	4	−1	1	+1	2	−1	1	0.99	13.9
npb-cg	+1	2	−4	4	−9	9	−2	2	0.96	11.9
npb-is	0	1	−2	2	−5	5	−1	1	0.98	5.6
npb-mg	−2	2	−3	3	−6	6	−1	1	0.84	22.4

to 1.04× on 20 of the 23 programs. The plain RSS run includes the 40 ms that the external sampler takes to start, which explains the ratios of 0.60× for atax and 0.84× for NPB-MG, whose runs last 0.10 s and 0.31 s. darknet runs 1.21× slower under the model in all 3 runs. A thread that sleeps for 0.99 s between samples slows darknet by the same 1.22×, so the cost comes from the presence of a second thread in a single-threaded program, and not from the sampling. We did not identify which code path the second thread slows down. The mode that reads `smaps` takes 0.98× the time of a plain RSS run at the median (0.60× to 1.18×). MemPrint's W5 run with its counter run takes 12.0× a native run at the median, 8.9× on the other programs and 34.4× on the allocator-heavy programs, with 93.0× on `cc1`.

5.1 Where the remaining RSS goes

Table 3 shows the size of each part of the model's run against the truth. On the allocator-heavy programs, glibc holds free memory equal to 6% to 39% of the truth on 5 of the 7 programs, and file-backed pages make up 6% to 88% of the truth. On the other programs, both parts are small, at a median of 1% and 6%.

miniVite has the largest share of both parts. glibc's free bytes amount to 39% of the truth, and file-backed pages amount to 88%. A sample of `/proc/pid/smaps` in the middle of a run shows

Table 3: Parts of the model's run, each summed over the run as a percentage of the truth's sum (the truth is 100). Data is estimate 1. Free is glibc's free bytes. File RSS and file PSS are the RSS and PSS of file-backed pages (RSS minus anonymous RSS, and PSS minus anonymous RSS). Median of 3 runs.

Program	Data (%)	Free (%)	File RSS (%)	File PSS (%)
miniVite	93	39	88	57
python	97	15	20	9
lua	99	23	6	1
sqlite	101	6	9	4
perl	91	15	10	4
quicksilver	96	2	23	15
cc1	95	2	23	20
others (median of 16)	99	1	6	0

where the file-backed pages are. The MPI library holds 2348 KiB, PMIx 2096 KiB, the C library 1496 KiB and the C++ runtime 1356 KiB, against a truth of 9.7 MiB on average over the run. The PSS of the C library is 77 KiB and that of the C++ runtime 199 KiB, because other processes map the same pages, but the PSS of the MPI and PMIx libraries equals their RSS, because only miniVite maps them. Subtracting glibc's free bytes alone therefore reduces miniVite's error only from +105% to +66% (Section 10), and leaving out the libraries' pages reduces it to −7%.

6 Transfer to a larger input

A scheduler rarely has a profile of the job's own input, so a footprint model is more useful if a curve measured on small inputs predicts the curve of a larger one. We apply one transfer rule, fixed before looking at the results, to every estimate. The peak at the largest input is a linear regression of the smaller inputs' peaks on the input size [2], never below the largest peak seen, and the shape is the curve of the largest smaller input, scaled to that peak. The largest input is held out, and the model's smaller inputs run once each.

The data estimate transfers as accurately as MemPrint (Table 4). On the allocator-heavy programs, its median whole-run error is −4% (10% absolute), against +20% (22%) for RSS and −6% (10%) for MemPrint. On the other programs, it is −2% (6%), against +2% (10%) for RSS and 0% (5%) for MemPrint.

Three cases show where transfer fails. On `cc1`, every estimate is 226% to 351% too high, because the three inputs are different source files whose size predicts the compiler's memory poorly. On NPB-MG, the data estimate is 67% too low and RSS 96% too low, because the only usable smaller input runs for a few milliseconds and the 5 ms sampling misses its peak, while MemPrint counts every reference and is 15% too high. On 2mm, gemm and jacobi-2d, RSS is 25% to 35% too low for the same reason, because the smallest input's run fits in one sample of the external sampler, which sees only the runtime. The model's thread starts before `main` and samples these runs, and its error on them is −10% to −4%. The transfer of a model trained on short runs therefore depends on sampling the start of a run, and sampling faster during the first milliseconds of a run would address it.

Table 4: Whole-run error (% of the truth), signed and absolute, of each estimate predicted for a program's largest input from its smaller inputs. The peak is a linear regression of the smaller inputs' peaks on the input size, never below the largest of them, and the shape is the largest smaller input's curve. – marks a program whose smaller inputs show nothing above the runtime's pages, for which the scheduler falls back to the user's request.

	RSS		Data (1)		Shared (2)		MemPrint	
Program	sign.	abs.	sign.	abs.	sign.	abs.	sign.	abs.
Allocator-heavy								
miniVite	+118	119	+36	45	+102	104	−6	7
python	+43	43	0	10	+6	11	+2	10
lua	+20	22	−4	6	−4	6	−17	17
sqlite	+12	20	−4	20	−2	19	−2	9
perl	+19	20	−10	10	−7	8	−14	14
quicksilver	+16	21	−6	6	−2	2	−7	7
cc1	+226	227	+248	248	+313	314	+351	351
Others								
2mm	−27	27	−4	8	−1	11	+4	5
atax	–	–	–	–	–	–	+21	21
gap-bfs	+6	10	+1	5	+1	6	−4	6
gap-cc	+5	8	−1	5	−1	5	−5	5
gap-pr	+5	16	−3	19	−3	19	−4	14
gap-sssp	+7	10	+2	6	+1	6	−1	4
gemm	−35	35	−10	11	−7	14	+2	2
hpccg	−1	1	0	1	0	1	−1	1
jacobi-2d	−25	25	−5	7	0	12	0	0
kripke	0	2	0	2	0	2	0	1
lulesh	+5	9	+3	8	+3	8	0	6
minife	+4	6	0	2	+2	5	−2	2
npb-cg	+5	8	−2	2	−2	2	+4	5
npb-is	−8	17	−18	19	−18	19	−14	15
npb-mg	−96	97	−67	68	–	–	+15	28
median, allocator-heavy	+20	22	−4	10	−2	11	−6	10
median, others	+2	10	−2	6	0	6	0	5

7 Scheduling with the model's curves

We place the model's curves in DMSched-Sim, the trace-driven scheduling simulator of the corpus. The simulator replays a synthetic job stream with MetaCentrum's shape (3000 jobs, 1 to 16 cores, run times of 5 to 720 minutes) on 32 nodes with 32 cores and 128 GB each and a 1 TB memory pool. Each job takes the footprint curve of a random corpus program with a smaller input, scaled to the job's peak, and the scheduler reserves memory by the oracle's estimate of that curve. The reservation divides the run into 20 windows, holds the estimate's largest value in each window, widens each window to its neighbours, and adds a margin times the peak. A job whose memory exceeds its reservation is killed, loses its work, and is requeued with a reservation 1.5 times larger. Each oracle's margin is calibrated by leaving one program out, so a job of program p gets the 90th percentile of the margins the other programs need to never fall below their truth.

On the pages truth (Table 5), the data estimate gives the lowest mean response time of all estimates that profile the job's own input, 433 minutes at a load of 0.8, against 515 for the RSS

Table 5: Scheduling simulation under the pages-touched truth on the native clock: mean response time (minutes) at three offered loads, and throughput (jobs per hour), OOM kills and core-hours lost to them at a load of 0.8. Means over 3 seeds of 3000 jobs on 32 nodes. Every estimate's margin is the 90th percentile of the margins the other programs need (leave one program out), and the user's request takes 10%. The second block profiles the job's own input, and the third transfers from the program's smaller inputs.

	Mean response (min)			At load 0.8		
Oracle	0.6	0.8	1.0	jobs/h	kills	lost core-h
Exact curve (truth)	55	94	261	62.3	0	0
Exact peak	172	479	698	47.4	0	0
User request	1867	2244	2452	23.6	0	0
KS+	491	976	1233	33.7	9236	5012
RSS peak	532	875	1126	35.4	412	886
RSS curve	185	515	776	41.3	407	1732
MemPrint W5	1247	1607	1797	27.2	386	1363
MemPrint W5, native clock	197	519	707	42.6	384	593
Data (1)	159	433	644	44.9	319	1888
Shared (2)	192	508	691	41.7	332	1916
RSS peak	1913	2269	2476	22.3	418	7
RSS curve	2369	2632	2869	20.3	698	553
MemPrint W5	1890	2273	2475	22.3	387	1363
MemPrint W5, native clock	911	1283	1477	30.5	275	699
Data (1)	2523	2806	3142	20.4	926	570
Shared (2)	2017	2479	2665	21.0	939	1554

curve, 519 for MemPrint on the native clock and 479 for the exact peak. It also has the highest throughput of these estimates (44.9 jobs per hour) and the fewest kills (319). Its calibrated margin is 7%, against 12% for MemPrint, because no program needs a margin above 9% with it. MemPrint on Pin's own clock gives 1607 minutes, because its curve and the job's footprint disagree on when memory is used.

Every transferred estimate except MemPrint on Pin's clock gives a response time at least twice that of the same estimate on the job's own input. MemPrint on the native clock transfers best (1283 minutes), and the data estimate transfers worst (2806 minutes), although its median transfer error equals MemPrint's (Section 6). The calibrated margin follows the worst programs, and the data estimate needs a margin of 1.00 on gemm and 2.67 on NPB-MG, the programs whose smaller inputs are too short for 5 ms sampling, so its 90th percentile is 0.53, against 0.23 for MemPrint.

Under the RSS truth, which the kernel charges and OOM-kills on, the RSS curve of an earlier run is the best estimate (393 minutes at a load of 0.8), because it measures the truth itself. The data estimate gives 748 minutes, close to MemPrint on the native clock (699 minutes), because it leaves out the retained memory and libraries that the RSS truth counts, and its calibrated margin must cover them. The estimate a scheduler should use therefore depends on what the node enforces. A limit on RSS calls for the RSS curve, and a limit on the memory the job uses, as in a cluster whose allocators return freed memory, calls for the data estimate.

8 Other allocators and transparent huge pages

We ran the 7 allocator-heavy programs under jemalloc 5 and TCMalloc (gperftools 2.18, `tcmalloc_minimal`), loaded with `LD_PRELOAD`, and measured the truth with Pin under the same

Table 6: Signed whole-run error (% of the truth under the same allocator) under jemalloc and TCMalloc, with transparent huge pages enabled (the machine's setting) and disabled for the process. *Anon* is the data estimate without the allocator's free bytes. Median of 3 runs. Overhead is the run time over a plain RSS run under the same allocator.

	THP enabled				THP disabled		
Program	RSS	Anon	Data (1)	Shared (2)	RSS	Data (1)	Shared (2)
jemalloc							
miniVite	+187	+189	+149	+202	+83	−10	+27
python	+59	+61	+35	+43	+49	+5	+14
lua	+112	+125	+60	+61	+96	+10	+12
sqlite	+29	+49	+21	+23	+26	−4	−1
perl	+26	+39	+22	+26	+25	−1	+4
quicksilver	+56	+59	+58	+69	+23	+4	+16
cc1	+28	+19	+15	+35	+25	−2	+19
median	+56	+59	+35	+43	+26	−1	+14
TCMalloc							
miniVite	+76	+21	−16	+21	+68	−27	+10
python	+48	+32	−3	+6	+47	−5	+4
lua	+62	+55	0	+3	+63	0	+3
sqlite	+46	+39	−22	−18	+46	−24	−20
perl	+17	+10	−7	−3	+19	−8	−2
quicksilver	+23	+2	−2	+10	+21	−3	+10
cc1	+25	+2	−4	+16	+22	−6	+14
median	+46	+21	−4	+6	+46	−6	+4
overhead (×)	1.00	0.98	0.98	0.98	1.00	0.99	0.99

allocator (Table 6).

TCMalloc. RSS overestimates the footprint by a median of 46%, more than under glibc, because TCMalloc keeps freed spans in its page heap and its caches. The data estimate has a median error of −4%, and it is low on SQLite (−22%) and miniVite (−16%). Without the allocator's free bytes, the estimate is +21%, so the statistics remove most of the gap.

jemalloc. With transparent huge pages enabled, every estimate overestimates jemalloc's footprint, RSS by a median of 56% and the data estimate by 35%, with +149% on miniVite. jemalloc places its extents at 2 MiB aligned addresses, so the kernel backs them with huge pages, and a 2 MiB page counts in full when the program touches 4 KiB of it. In a run of Lua under jemalloc, 45 MiB of the 65 MiB of anonymous memory is in huge pages. We disabled transparent huge pages for the process with `prctl(PR_SET_THP_DISABLE)`, which the library calls in its constructor on request. The data estimate's median error under jemalloc then drops to −1%, and RSS's to +26%. Under TCMalloc and under glibc, disabling transparent huge pages changes the median errors by at most 4 points (Table 7).

The memory that huge pages add is real. A node with transparent huge pages enabled holds the whole 2 MiB page in memory, and the pages-touched truth undercounts it, as MemPrint would. A scheduler on such a node needs the estimate with huge pages, which the model measures and Pin does not, and a scheduler that wants the 4 KiB footprint can disable huge pages for the profiling run. We also checked two other kernel counters that we expected to matter. `LazyFree`, the pages released with `MADV_FREE` that stay resident until the kernel reclaims them, was 0 in the jemalloc

Table 7: The corpus under glibc with transparent huge pages enabled and disabled for the process: median signed and absolute whole-run error (% of the truth). The data estimate here comes from the mode that reads `smaps`.

	RSS				Data (1)			
	THP on		THP off		THP on		THP off	
Programs	sign.	abs.	sign.	abs.	sign.	abs.	sign.	abs.
Allocator-heavy (7)	+16	17	+15	15	−3	9	−7	9
Others (16)	+1	2	0	3	−1	3	−1	4

Table 8: Signed whole-run error (% of the pages-touched truth) on the language model engines, median of 3 runs. Data (1) without files is $\max(A - F, 0) + B$. The model's overhead is the run time of the mode that reads `smaps` over a plain RSS run, and MemPrint's is its W5 and counter runs over a native run (×).

		Data (1)				Data (1)	Overhead (×)	
Engine and input	RSS	no files	with files	Shared (2)	MemPrint	THP off	model	MemPrint
llama2c 110M	+12	−86	+12	+11	0	0	1.00	3.9
llama2c 15M	0	−92	0	−3	−2	0	0.98	6.0
llama2c 42M	+15	−81	+15	+14	0	−1	1.00	4.4
llamacpp 135m-n128	+1	−39	−1	0	−3	−7	0.99	38.9
llamacpp 135m-n32	−26	−56	−28	−27	−13	−30	0.94	67.4

runs, and the `Referenced` bytes of llama2.c's model file equal its RSS.

9 Language model engines

llama2.c maps its model file and reads the weights from the mapping, and llama.cpp maps its GGUF file and copies part of it into buffers it allocates. Without the data-file term, the data estimate is 81% to 92% too low on llama2.c and 39% to 56% too low on llama.cpp (Table 8), because the weights are file-backed pages. With the term, the data estimate is within −1% to +15% of the truth on llama2.c and on llama.cpp generating 128 tokens. On llama2.c, the remaining overestimate comes from transparent huge pages. The engine allocates its key and value caches for the whole context of 1024 positions with `calloc` and fills only the positions it generates, and 45 MiB of its 47.5 MiB of anonymous memory is in huge pages. With huge pages disabled, the data estimate is within 1% of the truth on all three model sizes.

llama.cpp generating 32 tokens is the exception. Every estimate of the model is 26% to 28% too low, against 13% for MemPrint. The run lasts less than a second, and most of it loads the model, so the estimates and the truth disagree on when the model's pages become resident. The model's run costs 0.94× to 1.00× a plain RSS run, and MemPrint's W5 costs 3.9× to 6.0× on llama2.c and 38.9× to 67.4× on llama.cpp.

10 Other ways to correct RSS

Before we settled on the model, we tested four alternatives on the allocator-heavy programs under glibc, each on its own (Table 9). The first, RSS minus glibc's free bytes, is the first half of the model.

Table 9: Signed whole-run error (% of the truth) of each alternative used alone on the allocator-heavy programs, median of 3 runs, and its overhead over a plain RSS run (×). Runtime statistics exist only for CPython and Lua. These runs keep the file-backed pages in every estimate.

Program	RSS	RSS−free	trim 5 ms	trim thr. 0	runtime stats	referenced
miniVite	+105	+66	+58	+55	–	−70
python	+25	+9	+11	+12	+5	+11
lua	+20	0	+2	+17	−19	+2
sqlite	+9	+4	+7	+9	–	−24
perl	+11	−3	+6	+7	–	−16
quicksilver	+17	+16	+14	+14	–	−55
cc1	+16	+15	+15	+15	–	−68
median signed	+17	+9	+11	+14	−7	−24
median absolute	18	10	11	15	35	36
overhead (×)	1.00	0.99	1.03	1.02	0.96	1.02

Releasing freed memory during the run. Instead of subtracting free bytes, the program can return them to the kernel, so that RSS drops when the program frees memory. We tested two ways. The first calls `malloc_trim(0)` every 5 ms from a thread in the process, which releases the free pages inside every arena with `madvise`. The second sets glibc's trim threshold to 0, which releases free memory at the top of the heap after every `free`. Both reduce the median error less than subtracting free bytes, +11% and +14% against +9%. A trim threshold of 0 releases only the top of the heap, which explains why it changes Lua's error by only 3 points. Both alternatives also change the program's allocation behaviour, and the trim thread costs up to 1.10× on Perl.

Language runtime statistics. For CPython, we sampled the free bytes of its object allocator (the arena bytes minus the bytes in allocated blocks, from `sys._debugmallocstats`) from a thread started in `sitecustomize`. For Lua, we sampled the heap size the garbage collector reports (`collectgarbage("count")`) from a debug hook every 10^6 virtual machine instructions. CPython's whole-run signed error drops to +5%, but its absolute error rises, which means that the estimate is too low in some parts of the run and too high in others. We did not determine why. Lua's heap count leaves out the interpreter's own memory and gives −19%. The statistics exist only for these two programs, and GCC reports its heap only at the end of a compilation.

Referenced pages per window. The kernel can clear the referenced bits of a process's pages (`/proc/pid/clear_refs`), after which `smaps_rollup` counts the pages the process has touched since. We cleared the bits 20 times per run and took the pages referenced in each window as the footprint of that window, in the manner of a working set [4]. The estimate falls below the truth by up to 70% (miniVite), 68% (`cc1`) and 55% (Quicksilver), because a page that the program holds but does not touch within a window is not counted. A scheduler that reserved this estimate would run these programs out of memory.

11 Limitations

A limitation of the free-byte correction is that it reads only the statistics of glibc, jemalloc and TCMalloc, which leaves the memory of other allocators in the estimate. The arenas of CPython's object allocator and GCC's garbage-collected heap stay in the estimate as RSS. On our corpus, the data estimate is nevertheless within 5% of the truth on CPython and `cc1`, and it is unclear whether this result holds for inputs on which these allocators retain more memory.

A limitation of the data-file rule is that it classifies mappings by path, which counts any mapped file that is not a shared object as data. A program that maps a large file it uses as code, or a library without `.so` in its name, would have that file counted.

A limitation of the free-byte correction is also that some free bytes may not be resident, such as the untouched part of the top of the heap, which makes the data estimate lower than the truth. The data estimate is below the truth on 6 of the 7 allocator-heavy programs under glibc, by 1% to 9% over the run, and on SQLite under TCMalloc by 22%, so a scheduler that uses it needs a margin.

A limitation of the sampling thread is that it adds a second thread to a single-threaded program, which slows darknet by 1.21× (Section 5). Sampling from an external process avoids the thread, but the allocator's statistics can be read only inside the process. A second limitation of the thread is its 5 ms interval, which misses the curve of runs shorter than a few tens of milliseconds and limits the model's transfer from the smallest inputs (Section 6 and Section 7).

The evaluation covers a single machine, short runs of 0.1 to 12 s and single-process programs. The truth's clock mapping passes through the native run's reference counts, which also place MemPrint's curve, so MemPrint's timing accuracy is overstated against the truth. The truths under jemalloc and TCMalloc use the clock of the native run under glibc.

12 Related work

Footprint measurement. Valgrind's Massif and DHAT [15, 14] and MemPrint measure the heap or the touched memory of every allocation by instrumenting the program. Our model reads counters that the kernel and the allocator already keep. Unlike MemPrint, it does not attribute memory to allocation sites or show which bytes of a block the program touches.

Page access sampling. DAMON [18] and Thermostat [1] sample the accessed bits of pages to find cold memory, and PEBS sampling [16] records the addresses of sampled loads. These methods measure which pages a program uses within a period of time, which is the working set [4] and not the footprint. Our referenced-page alternative (Section 10) shows that the working set of a window falls far below the footprint on allocator-heavy programs.

Memory prediction for scheduling. KS+ [2] predicts a workflow task's memory over time from the RSS of earlier runs, and Zacarias et al. [20, 21] predict the memory of HPC jobs from earlier runs. Pond [11] predicts how much of a virtual machine's memory goes untouched. These systems learn from RSS or from allocated memory, and our model gives them a training signal without the allocator's retained memory and the job's shared libraries at the same cost as RSS.

Allocators and huge pages. Mesh [19] compacts the heap to return fragmented memory to the kernel, and TCMalloc's Temeraire [8] releases free huge pages. Ingens [10] and HawkEye [17] limit the memory that huge pages waste on partly used ranges. These systems reduce the gap between RSS and the footprint by changing the system, whereas our model measures the gap without changing the program's allocations.

13 Conclusion

RSS overestimates the footprint of programs that allocate and free memory throughout their run, because the allocator keeps freed memory resident and because RSS counts shared libraries. A model that subtracts the allocator's free bytes from the anonymous RSS and adds mapped data files and the runtime's pages reduces the median whole-run error on these programs from +16% to −4%, which matches MemPrint's accuracy on the job's own input and on larger inputs, at a

median of 0.99× the run time of a plain RSS run instead of 34.4×. In a scheduling simulation, its curve gives the lowest response time of the estimates that profile the job's own input. Pin-based profiling remains needed for attributing memory to code, for allocators the model cannot query, for the curves of runs too short to sample, and for the 4 KiB footprint of programs whose memory the kernel backs with huge pages.

Artifacts. `paper_rss/tools/mprshim.c` is the sampling library. `run_model.sh`, `run_transfer.sh`, `run_options.sh`, `run_alloc.sh` and `run_llm.sh` run the experiments, `grid.py` builds the simulator's corpus and runs the scheduling grid, and `compute.py` produces every table. The runs are in `~/memprint-data/2026-10-26-rsscurve`.

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